

## COMBINATORICS ASSIGNMENT 2

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6. Prove that

$$P(n, k) = k! \binom{n}{k}.$$

*Proof.* First let's consider how we choose our  $k$ -permutations of  $[n]$ . We begin by choosing  $n$  elements, then  $n - 1$ , then all the way down to  $n - k + 1$ , so we see that

$$P(n, k) = n(n - 1)(n - 2) \cdots (n - k + 1) = \frac{n!}{(n - k)!}.$$

On the right-hand side, we are choosing  $k$ -elements from  $[n]$  in an unsorted fashion, but then we need to sort them in  $k$  ways which is  $k!$ , and we see that

$$k! \binom{n}{k} = k! \left( \frac{n!}{k!(n - k)!} \right) = \frac{n!}{(n - k)!}.$$

□

10. Prove that:

$$\sum_{k=0}^n r^{n-k} \binom{n}{k} = (r + 1)^n$$

*Proof.* We'll go ahead and consider the general case here for  $(x + y)^n$ . We expand as:

$$\underbrace{(x + y)(x + y) \cdots (x + y)}_{n\text{-terms}}$$

which if we multiply out, we get  $2^n$  terms each of which can be arranged as  $x^{n-k}y^k$  for  $k \in [n]$ . We obtain the coefficient for the  $x^{n-k}y^k$  term by choosing  $y$  in  $k$  of the  $n$  factors and  $x$  necessarily in the remaining  $n - k$  factors. Thus the coefficient on  $x^{n-k}y^k$  becomes the  $k$ -subsets of  $n$  factors or  $\binom{n}{k}$ . So we have

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

For the case in the problem we simply set  $x = r$  and  $y = 1$ , and we are done. □

**16.** Prove Vandermonde's Identity:

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$$

*Proof.* The left-hand side counts all the  $k$ -element subsets of  $[m+n]$ . The right-hand side does the same thing according to the number of elements chosen from  $[n]$ . Indeed, we can first choose  $k$  elements from  $[m]$  in  $\binom{m}{k}$  ways, then choose the remaining  $r-k$  elements from the set  $\{m+1, m+2, \dots, m+n\}$  in  $\binom{n}{r-k}$  ways.  $\square$

**19.** Prove the identity:

$$\binom{n+1}{r+1} = \sum_{k=r}^n \binom{k}{r}$$

*Proof.* Let's go ahead and expand out the right hand side here so we see that:

$$\binom{n+1}{r+1} = \binom{r}{r} + \binom{r+1}{r} + \binom{r+2}{r} + \dots + \binom{n}{r}$$

Now with regards to the identity in question, the left-hand side clearly counts all the  $r+1$ -element subsets of  $[n+1]$ . And if we separate the right-hand side into cases according to the largest element. That is, there are  $\binom{r}{r}$  subsets of  $[n+1]$  that have  $r+1$  elements whose largest element is  $r+1$ ; there are  $\binom{r+1}{r}$  subsets of  $[n+1]$  that have  $r+1$  elements whose largest element is  $r+2$  and so on. In general there are  $\binom{r+k}{r}$  subsets of  $[n+1]$  that have  $r+1$  elements whose largest element is  $r+k+1$  for all  $k \leq n-r$ .  $\square$

**20.** Prove the identity:

$$2^n - 1 = \sum_{k=0}^{n-1} 2^k$$

*Proof.* We begin by noticing that the left hand side counts the non-empty subsets of  $[n]$ . To see the count of the right-hand side we count by classifying all the subsets of  $A \subseteq [n]$  by each's largest element  $m \in A$  where  $m \in \{1, 2, \dots, n\}$ ,  $\max(A) = m$ . We notice that every element of  $A$  is chosen from  $[n]$ , and elements  $\{1, 2, \dots, m-1\}$  can be included or not independently. So we have that:

$$2^n - 1 = \sum_{m=1}^n 2^{m-1}.$$

A simple shift in indices  $k = m - 1$  then results in:

$$2^n - 1 = \sum_{k=0}^{n-1} 2^k.$$

$\square$

**22.** Prove the identity:

$$\binom{n}{k} \binom{k}{t} = \binom{n}{t} \binom{n-t}{k-t}$$

*Proof.* The left-hand side is the number of ways that we can choose  $S$  of  $k$ -elements from  $[n]$ , and then subsequently choose  $t$ -elements from  $S$ . Regarding counting the right-hand side we see that we are immediately choosing  $t$ -elements from our original set of  $[n]$ . But then we are left with a set of  $[n-t]$  from which we need to choose  $k-t$  elements. So it works.  $\square$

**24.** Prove the identity:

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{r+k} = \binom{m+n}{m+r}$$

*Proof.* Let  $A$  and  $B$  be disjoint sets with  $|A| = m$  and  $|B| = n$ . We want to find the number of ways to choose  $S \subseteq A \cup B$  with  $|S| = m+r$ , clearly  $\binom{m+n}{m+r}$ .

Now to find the right-hand side we count by splitting according to how many are chosen from  $A$ . For any such subset  $S$ , let  $k = |S \cap A|$ . We see the following

- $j \in \{0, \dots, m\}$ , and
- once  $j$  is fixed we must choose:
  - $j$ -elements from  $A$ :  $\binom{m}{j}$  ways, and
  - the remaining  $((m+r) - j)$ -elements from  $B$  in  $\binom{n}{(m+r)-j}$  ways.

So the total count follows as:

$$\sum_{j=0}^m \binom{m}{j} \binom{n}{(m+n)-j}$$

Now we re-index setting  $k = m - j$  yielding  $(m+n) - j = r + k$  with  $\binom{m}{j} = \binom{m}{m-j} = \binom{m}{k}$ , and sum becomes:

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{m+k}$$

and we are done.  $\square$

**26.** Prove the identity:

$$\sum_{k=0}^n k^2 \binom{n}{k} = n(n-1)2^{n-2} + n2^{n-1} = n(n+1)2^{n-2}$$

*Proof.* Begin with the binomial theorem for  $(x+1)^n$ , which follows as:

$$(x+1)^n = \sum_{k=0}^n \binom{n}{k} x^k$$

Now differentiate both sides, and we get

$$n(x+1)^{n-1} = \sum_{k=1}^n k \binom{n}{k} x^{k-1}$$

Now multiply both sides of the equation by  $x$  yielding:

$$nx(1+x)^{n-1} = \sum_{k=1}^n k \binom{n}{k} x^k$$

Differentiating again we get

$$n[(x+1)^{n-1} + (n-1)x(1+x)^{n-2}] = \sum_{k=1}^n k^2 \binom{n}{k} x^{k-1}$$

Now if we substitute in  $x = 1$  we get

$$\begin{aligned} n[(2)^{n-1} + (n-1)x(2)^{n-2}] &= \sum_{k=1}^n k^2 \binom{n}{k} \\ n(n+1)2^{n-2} &= \sum_{k=1}^n k^2 \binom{n}{k} \end{aligned}$$

□

**33.** Prove the identity:

$$\sum_{k=n-s}^{m-r} \binom{m-k}{r} \binom{n+k}{s} = \binom{m+n+1}{r+s+1}$$

*Solution.*

□